

## EMILY DEEB

Geospatial Data Scientist | Urban Analytics | MSc Urban Spatial Science, CASA UCL

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## PROFESSIONAL SUMMARY

Geospatial data scientist and urban analyst with five years' experience in GIS (Geographic Information Systems) and urban analytics, including city-scale geospatial analysis, statistical modelling and data analysis. Through an industry dissertation with CACI Ltd, built Python-based machine learning (ML) pipelines on proprietary consumer spending data for national retail spending analysis. Skilled in building data pipelines, data cleaning, predictive modelling, stakeholder communication, and translating spatial findings into actionable insights. Seeking opportunities in spatial data science, urban analytics, or location intelligence.

## PROFESSIONAL EXPERIENCE

**CACI Ltd x GeoDS Programme, London** | May 2025 – Sep 2026

Industry Dissertation — Spatial Data Science

- Investigated whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition, using CACI proprietary consumer spending datasets and HiMoC morphometric classifications.
- Built an end-to-end Python-based machine learning pipeline using GeoPandas, scikit-learn, SHAP, and spatial autocorrelation analysis; applied pre-modelling diagnostics, feature engineering, model evaluation and statistical modelling to produce reproducible, documented analytical outputs.
- Found that physical form explained 30.1% of predictive variance beyond demographics across all six retail spending categories; identified street-network betweenness centrality as the strongest single physical predictor ( $r \approx 0.33$ ).
- Communicated findings through data visualisation and written reporting to CACI's Consumer and Location Intelligence team; presented at GeoDS Data Partner Forum, Saïd Business School, Oxford.

**Buro Happold, London** | Jun 2026

Industry Consultant — Building a Better Future Scheme

- Led research scoping and team coordination across a multidisciplinary group of six; managed stakeholder expectations, synthesised meeting outputs, and contributed to the final written report and client presentation delivered to Buro Happold and industry stakeholders.

**Derman Verbakel Architecture, Tel Aviv** | Sep 2020 – Sep 2025

Team Leader — Urban Design & GIS

- Led a team of 4 architects across 15 concurrent projects spanning block to city scale for municipal and private clients including Eilat Municipality and the Israeli Authority for Urban Renewal.
- Developed and led geo-computational workflows in QGIS and ArcGIS for geospatial analysis, morphological mapping, site feasibility analysis, market analysis, and statistical profitability modelling; improving office efficiency by 30%.
- Independently collected, cleaned, integrated, and aggregated multi-source spatial datasets from scratch for Sakhnin city centre regeneration; delivered evidence-based spatial reports and cartographic visualisations to municipal planning committees.
- Trained 15+ team members in GIS tools; developed reproducible, documented spatial workflows adopted across the office.

- Key projects: Transportation Masterplan of Eilat; Regeneration of Sakhnin City Centre (1,200 units); TOD Regeneration Plan Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).

## SELECTED PROJECTS

### London Crime Density & Urban Amenities Analysis | 2025

Python, Machine Learning, Spatial Analysis, London Datastore

- Analysed 1.78 million Metropolitan Police Service crime records across 679 London wards to investigate relationships between urban amenity distribution and crime density.
- Applied multiple regression, K-Nearest Neighbours (KNN), and Random Forest models with SHAP decomposition; found nightlife amenities strongest predictors of person/society crime, retail nodes driving property crime.

### Food Shock — Global Food Supply Crisis Tracker | 2026

Group Project — React, Leaflet, FAO Data, Python

- Built an interactive web application visualising global food supply chain disruptions using FAO trade flow data (200 countries, 82 commodities), commodity price indices, and spatial mapping.

## EDUCATION

### MSc Urban Spatial Data Science | Sep 2025 – Oct 2026

The Centre for Advanced Spatial Analysis (CASA), UCL

- Specialised in computational urban analysis, spatial data science, geocomputation, statistical modelling, spatial statistics, urban simulation, data visualisation, and machine learning in Python and R.

### B.Arch. Architecture | Sep 2015 – Jun 2020

Bezalel Academy of Arts and Design Jerusalem

- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation. Nominated for David Azrieli Architecture Prize.

## TECHNICAL SKILLS

**Programming & Data:** Python (Pandas, NumPy, Matplotlib, GeoPandas, Scikit-learn), R (RStudio), SQL, PostgreSQL, React, Leaflet.

**Data Science & ML:** Machine Learning (ML), Classification, Clustering, Regression, Random Forest, KNN, Feature Engineering, Model Evaluation, Statistical Analysis, Spatial Statistics, SHAP, Spatial Autocorrelation, Pre-modelling Diagnostics, Predictive Modelling.

**Data & Viz:** Data Pipelines, Data Cleaning, Reproducible Workflows, Data Visualisation, Dashboards, Power BI, Quarto, Jupyter, Git.

**Spatial & GIS:** GIS (Geographic Information Systems), Geospatial Analysis, QGIS, ArcGIS, Spatial Network Analysis, Spatial Data Science, Urban Morphology, Cartography, Google Earth Engine API, Remote Sensing.

**Design & Other:** AutoCAD, Rhinoceros, Adobe Suite (Photoshop, Illustrator, InDesign), Microsoft Office (Excel, Word, PowerPoint), UI/UX (CalArts certified).

## LANGUAGES & CERTIFICATIONS

English (fluent), Hebrew (fluent), Arabic (native) | Licensed Architect — Israeli Labor Department, 2025.

**UK right to work:** Graduate visa (valid until 2029), no sponsorship required.